

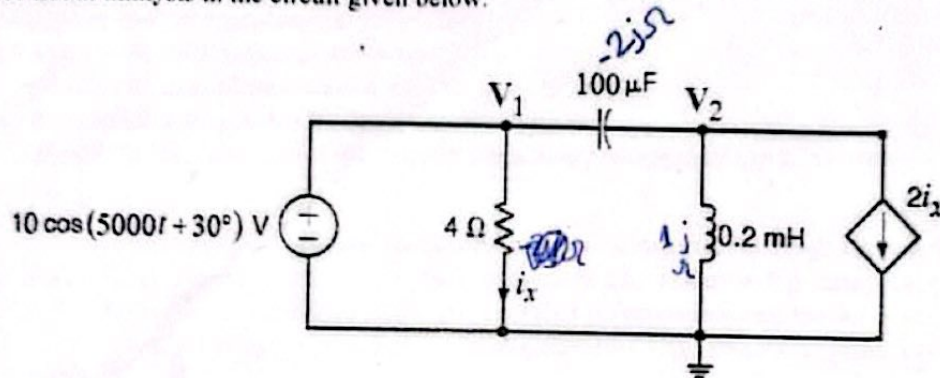
	Course Name:	Electrical Network Analysis	Course Code:	EE-2004
	Program:	BS(EE)	Semester:	Fall 2023
	Duration:	60 Minutes	Total Marks:	30
	Paper Date:	02/10/2023	Weight:	15%
	Section:	All	Page(s):	1
	Exam Type:	Midterm -1	CLO	1

Student : Name: _____ Roll No. _____ Section: _____

Instruction/Notes: • Attempt all the questions and solve all parts of a question together
 • Carefully understand the question before attempting
 • Show all calculations, direct answers are not acceptable

Question # 1 CLO 1 [3 + 10 + 2 = 15 marks]

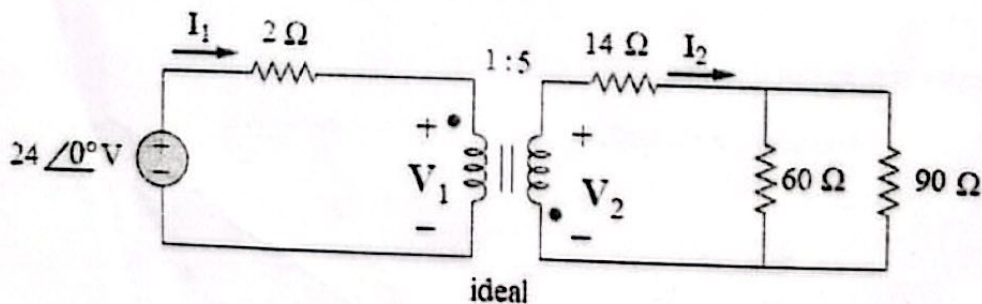
Use nodal analysis in the circuit given below:



- Draw the frequency domain equivalent circuit.
- Determine the node voltages, V_1 and V_2 .
- Calculate the value of i_x and determine its steady state expression.

Question # 2 CLO 1 [5 + 10 = 15 marks]

Solve the ideal transformer circuit given below to:



- Calculate the impedance reflected into the primary winding of the transformer
- Determine I_1 , I_2 , V_1 and V_2